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SQL Server 2025 Meets Microsoft Fabric: Seamless Integration for Modern Data Solutions

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Hey!

Jeg er Magi

Jeg er her fordi jeg digger
MVP- Dagen

Du finner meg på magi@inspir-it.no



► Agenda today

- ▶ Intro and data integration concepts
- ▶ ETL batch processing integration
- ▶ Fabric Mirroring for SQL Server OnPrem
- ▶ Other Integration options for SQL Server for Cloud Solutions



0

Intro and data integration concepts

Let's put the stage for integration



SQL Server 2025

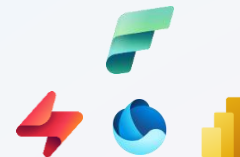
The AI-ready enterprise database from ground to cloud



AI built-in



Develop modern
data applications



Integrate your data
with Fabric



Secure by default



Mission critical engine



Connected
with Arc



Assisted by
Copilots



Benchmark
leader



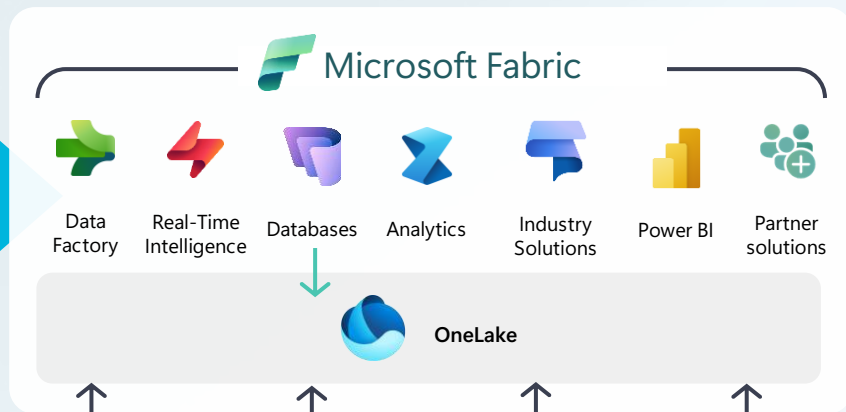
Optimized for
latest hardware



Built for all
platforms

Integrated with Microsoft Fabric

Integrate your operational data with a unified data platform



Data is transferred or replicated into One Lake and **kept up-to-date or near real-time**

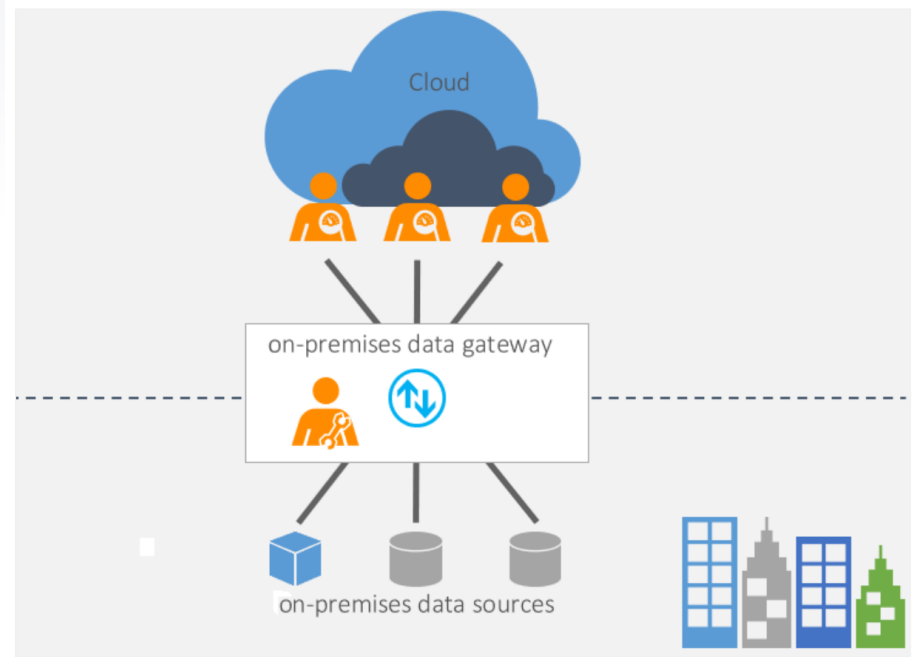
Batch transfer for ETL for DWH or Medallion Architecture

Mirroring **replicates databases to Fabric with zero ETL**

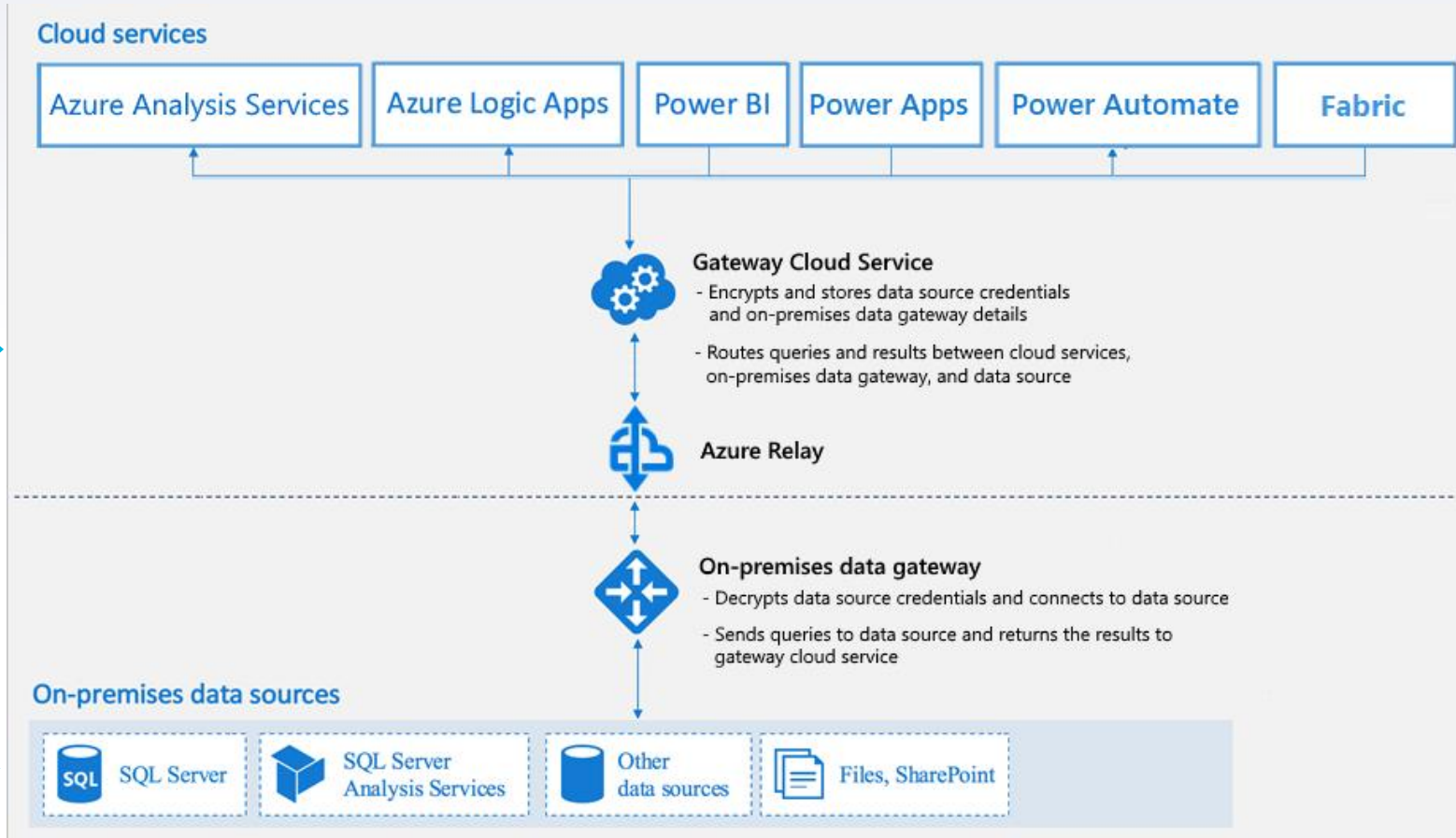
Secure and scalable transfer of data from OnPrem SQL Server to One Lake

On Prem Data Gateway – the foundation of SQL Server to MS Fabric Integration

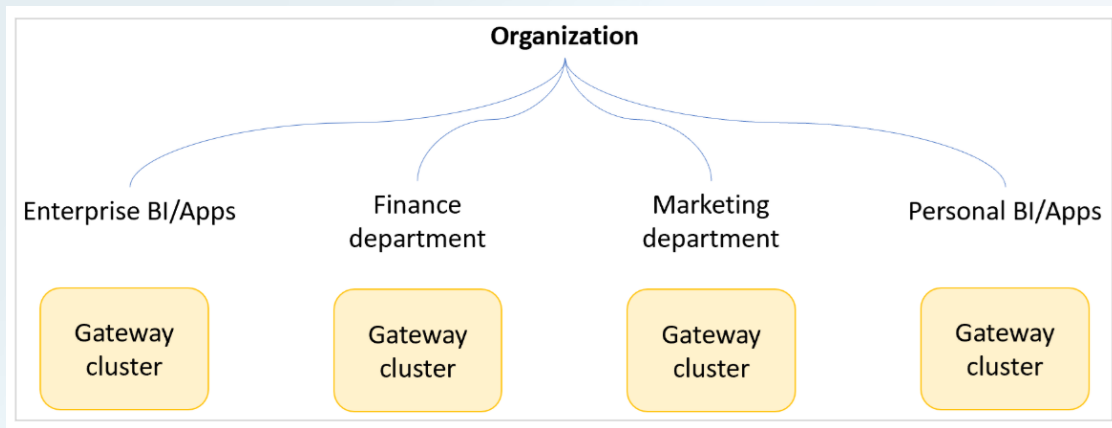
- Facilitates quick and secure behind-the-scenes communication.
- Initiates outbound connections to the cloud
- Requires no inbound connections



How the gateway works



Important consideration for OPDG implementations



- Plan and maintain gateway cluster scalability
- Use multiple gateway clusters
- Use the gateway high availability and load balancing features

1

ETL batch processing integration



Semi-S

DEMO



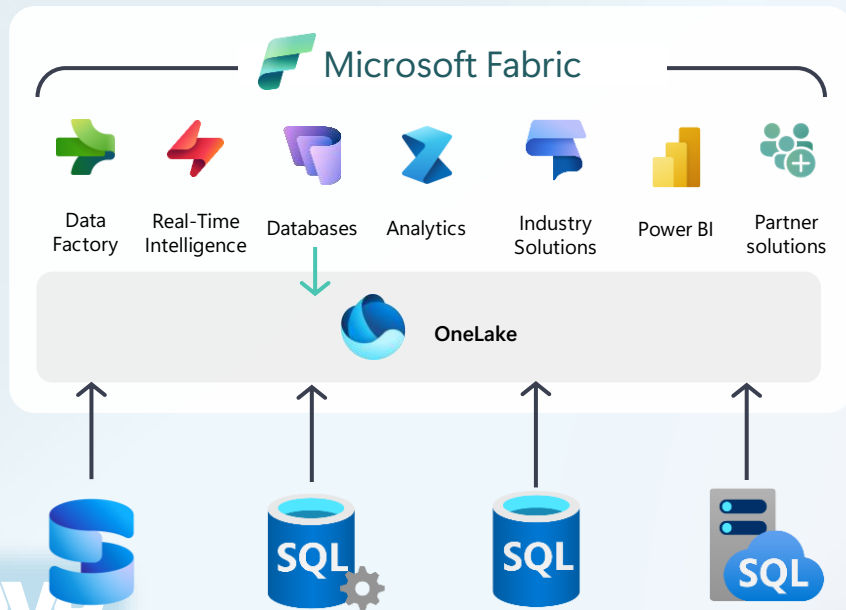
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Mirroring in Fabric



Microsoft Fabric Mirroring

Integrate your operational data with a unified data platform



Mirroring **replicates** databases to Fabric with **zero ETL**

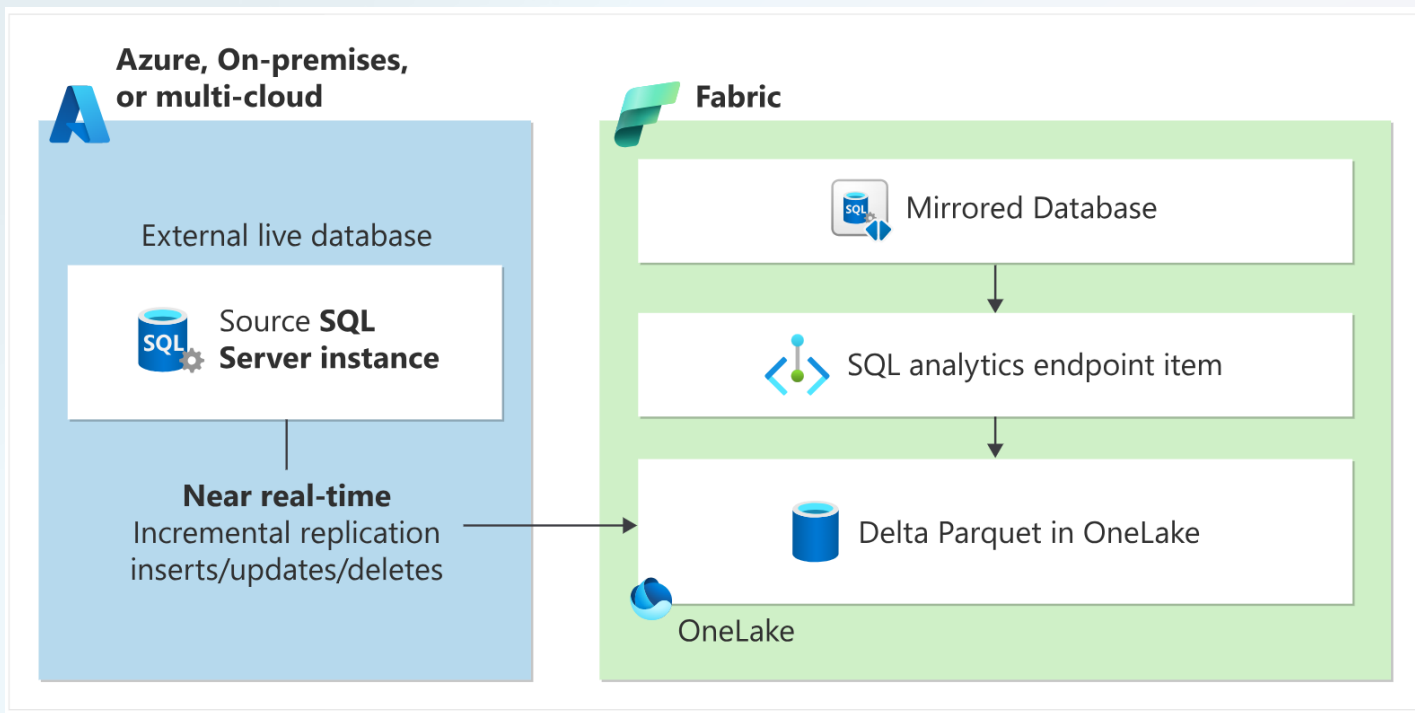
Data is replicated into One Lake and **kept up-to-date in near real-time**

Mirroring **protects** operational databases from **analytical** queries

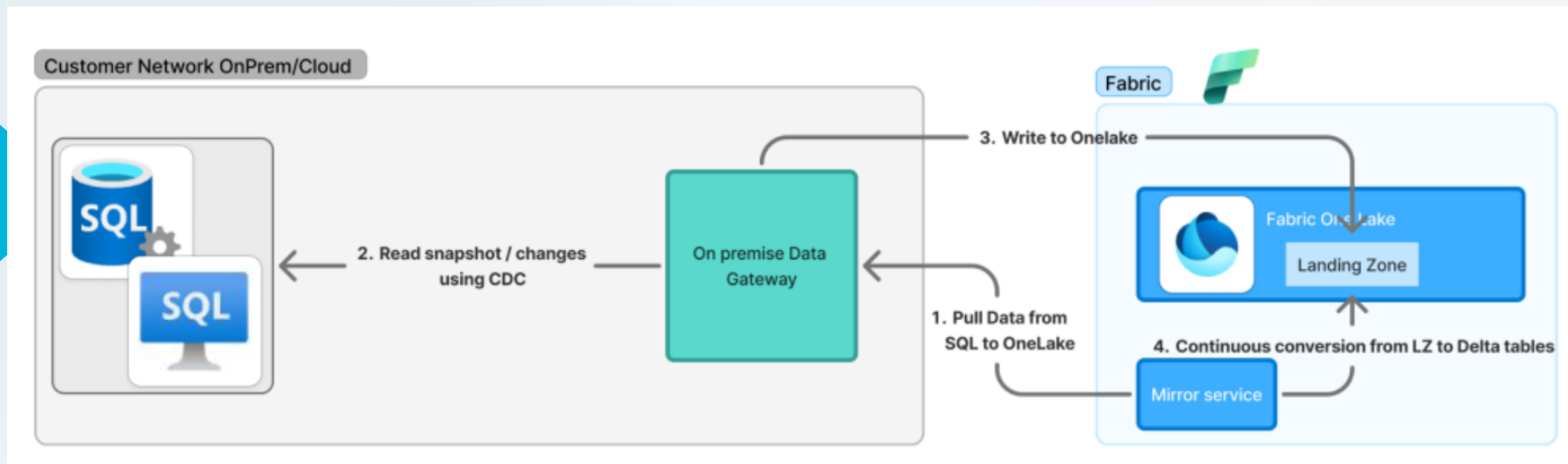
Compute replication is included with your Fabric capacity **for no cost**.

Free Mirroring Storage for Replicas tiered to Fabric Capacity.

Mirroring Concept



Fabric mirroring SQL Server 2016 - 2022



DEMO





Home



Workspaces



Copilot

OneLake
catalog

Monitor



Real-Time



Workloads

MVPDagen
2025My
workspace

Fabric



MVPDagen2025



Create deployment pipeline



Create app



Manage access



Workspace settings

+ New item

New folder

→ Import

Migrate

Filter by keyword

Filter



Name

Type

Task

Owner

Refreshed

Next
refresh

Endorsemer

Sensitivity

Included
in app

AWHS

Warehouse

—

Margarita ...

—

—

—

—



AWLKH

Lakehouse

—

Margarita ...

—

—

—

—



AWLKH

SQL analyt...

—

Margarita ...

—

—

—

—



copyjob1



Copy job

—

Margarita ...

—

—

—

—



ETLPipelineDemo_LKH

Pipeline

—

Margarita ...

—

—

—

—



ETLPipelineDemo_WHS

Pipeline

—

Margarita ...

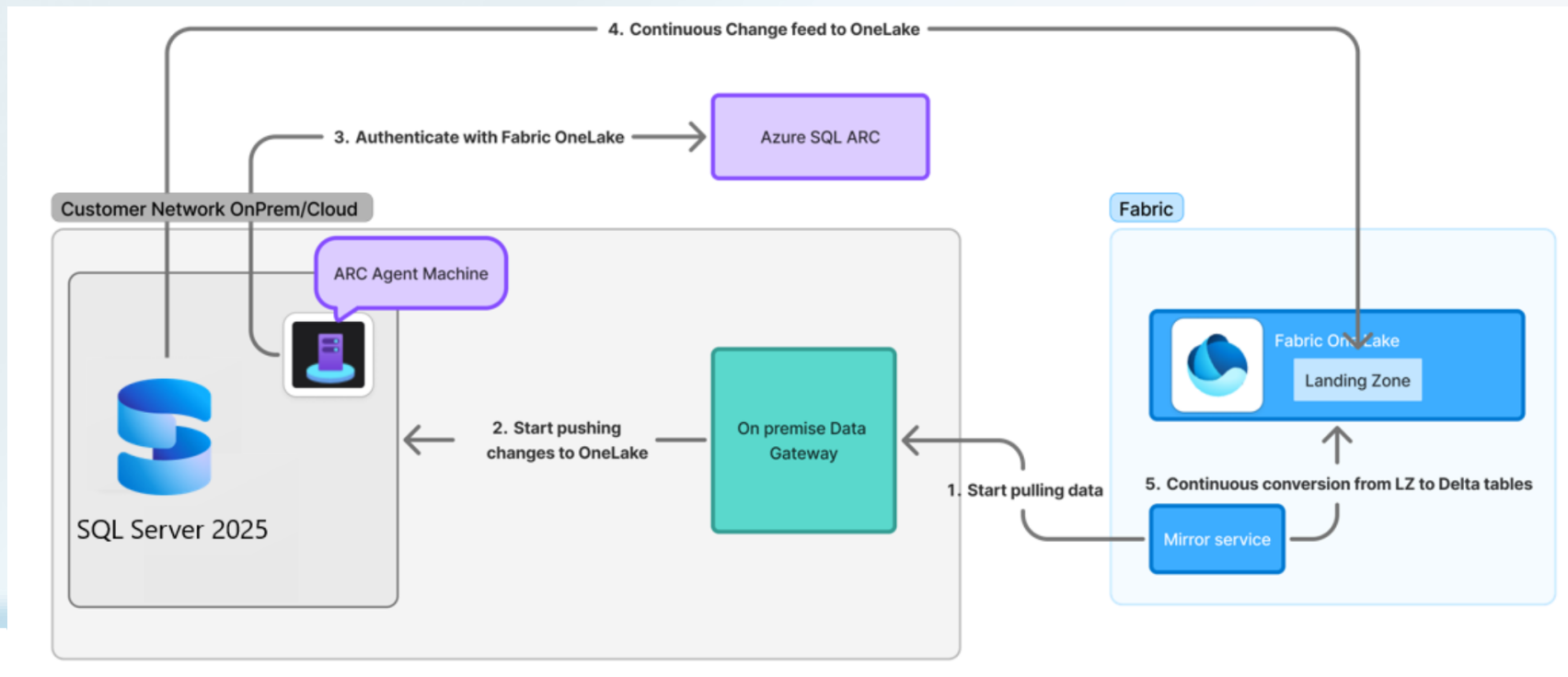
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Fabric mirroring SQL Server 2025

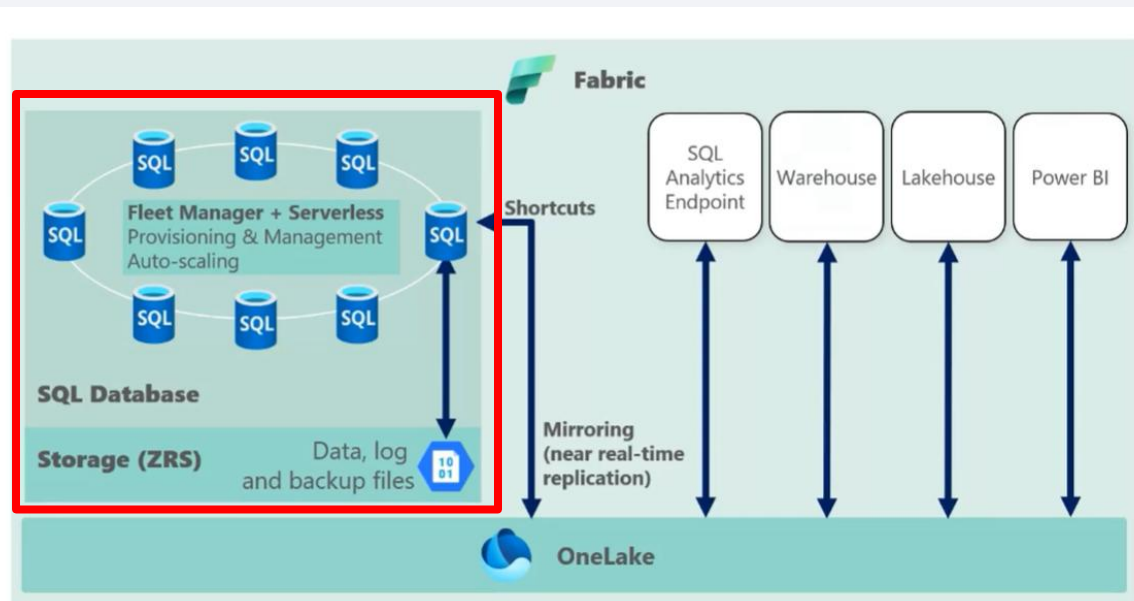


Capturing changes with SQL Server

	Change Data Capture (CDC)	Change Event Streaming (CES)
Capture Source	Committed transactions from transaction log (async). Requires SQL Server Agent	Committed transactions from transaction log (async)
Capture Destination	System table (extra logging required)	Event stream ("IOless")
Data Captured	Row identifier and data	Row identifier and data (DDL allowed)
Capture Method	Query system tables. Multiple destinations	Process event stream ("IOless"). Single destination
Availability	Near real-time (pull-based)	Near real-time (push-based)
Typical Use Cases	Data warehousing and auditing	Event-driven architectures – microservices integration, real-time analytics, cache sync, AI Agents

Ingest/ETL or Mirror?

- For frequent analysis when the source systems shouldn't be directly queries
- IF ETL is still needed to build a star schema, you can use stored proc over SQL Endpoint, but why mirror on the first place?
- Main use case for mirroring: Operational reporting & WH



Snowflake

MVP
Dagen
Mirroring for
Snowflake



Mirroring for Azure
SQL DB



Mirroring for
Cosmos DB



Mirroring for Azure
SQL MI



Mirrored Azure Databricks
Catalog

Extending the Integration

- SQL Server 2025 CES for Event based Solutions
- SQL Server 2019 – 2025 External tables for data archival





10



Response	Percentage
Yes, the current system is the best	65%
No, the current system is not the best	35%

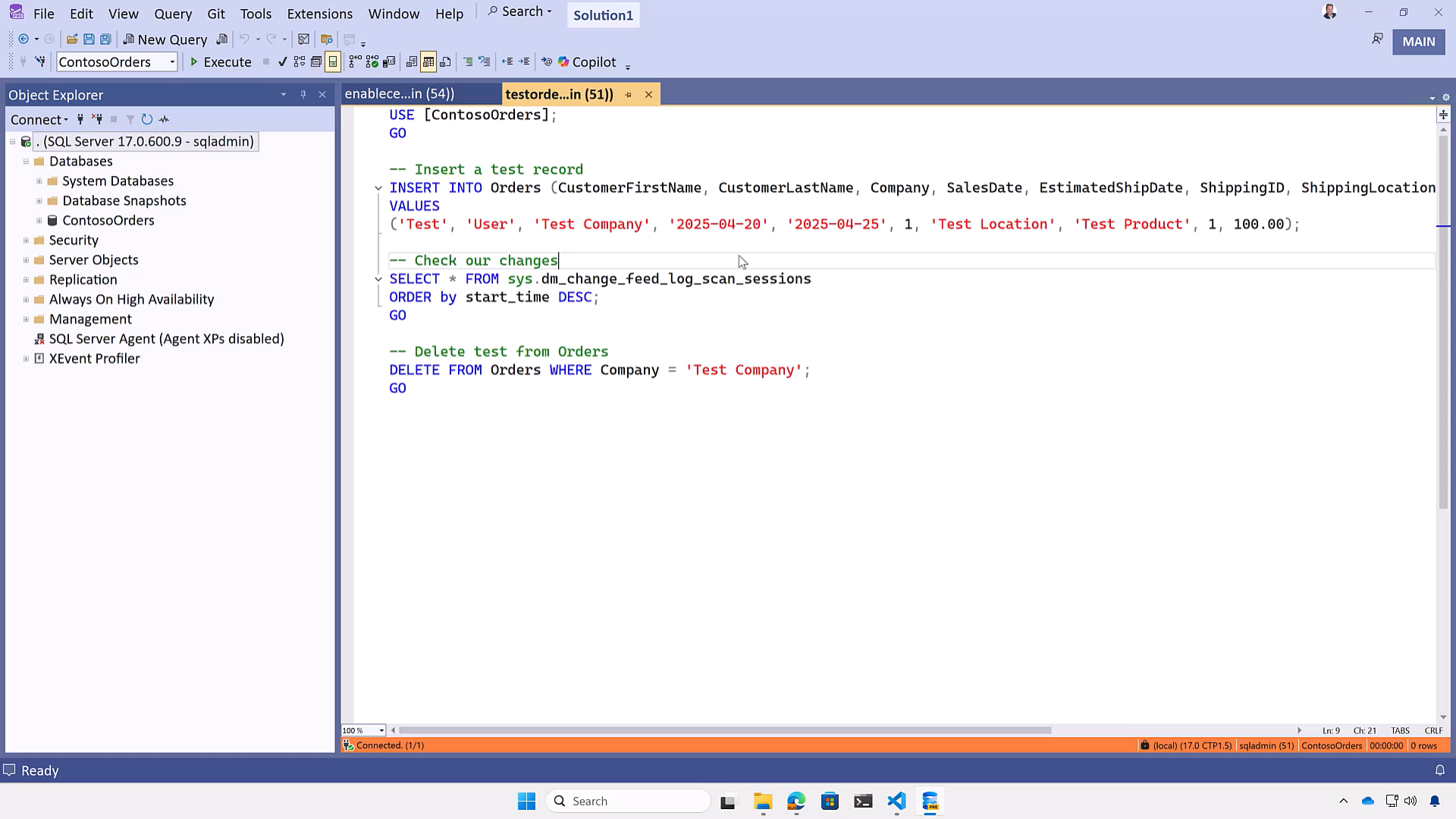
Ln: 24 Ch: 1 SPC CRLF

100

be

tic
ce

🔒 (local) (17.0 CTP1.5)	sqladmin (54)	ContosoOrders	00:00:00	2 rows
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sqllevntsfunctionapp Function App

Search

- Browse Refresh Stop Restart Swap Get publish profile Reset publish profile Download app content Delete Send us your feedback

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Microsoft Defender for Cloud
- Events (preview)
- Recommended services (preview)
- Resource visualizer
- Functions
- App keys
- Proxies
- Deployment
- Deployment Center
- Settings
- Environment variables
- Configuration
- Deployment settings
- Authentication
- Identity
- Custom domains
- Certificates
- Networking

Essentials

View Cost JSON View

Resource group (move)	: sqllevnthubrg	Default domain	: sqllevntsfunctionapp.azurewebsites.net
Status	: Running	Operating System	: Linux
Location (move)	: South Central US	Plan type	: Flex Consumption
Subscription (move)	: AzureSQL bobward	Instance Memory	: 2048 MB
Subscription ID	: 0efc44aa-c965-420f-aac4-fff305dbcc97		
Tags (edit)	: Add tags		

- Functions Properties Notifications (0)

Create functions in your preferred environment



Create with Visual Studio

- Best optimized for:
- Local development within Visual Studio
 - Using a full .NET IDE

Create with Visual Studio



VS Code Desktop

- Best optimized for:
- Local development within VS Code
 - Custom development tool requirements

Create with VS Code Desktop



Other editors or CLI

- Best optimized for:
- Using preferred editor for development
 - Visual Studio, IntelliJ, command line

Set up your editor

SQL Server External Tables to ADL

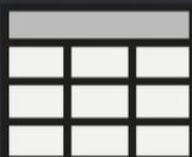
ARCHIVING COLD DATA TO A LAKE



Partitioned Table



Partition
Switch

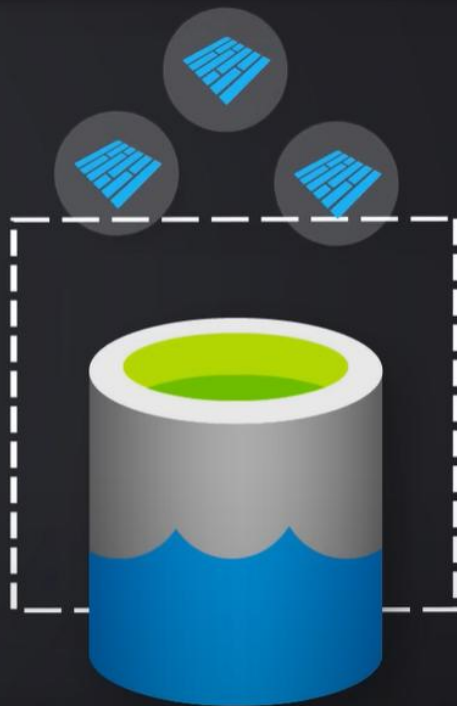


Archive Table
with cold data



CETAS

(Create External
Table as Select)



Using archived data from the ADL



Takk!

Spørsmål?

Du finner meg på

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